

Chapter 4 — Loops

Original educational notes prepared as sample source material for the RAG assignment.

Why Use Loops?

Loops repeat a block of code. They are useful when the same operation must be performed for multiple items or repeated until a condition changes.

for Loop

A for loop iterates over items in an iterable such as a list, tuple, string, or range. Example: for number in range(5): followed by an indented print(number).

while Loop

A while loop repeats while its condition remains True. The condition should eventually become false to prevent an unintended infinite loop.

break and continue

The break statement immediately exits a loop. The continue statement skips the remaining statements in the current iteration and continues with the next iteration.